
SmoIVLM

Redefining Small and Efficient Multimodal Models

Compact Vision-Language Models for Edge Deployment

Hugging Face & Stanford University

256M
Parameters

<1GB
GPU RAM

Open Source
Weights, Data, Code

Large VLMs Are Powerful but Impractical for Edge Deployment

State-of-the-art VLMs demand 27+ GB RAM — over 30× what mobile devices can spare.

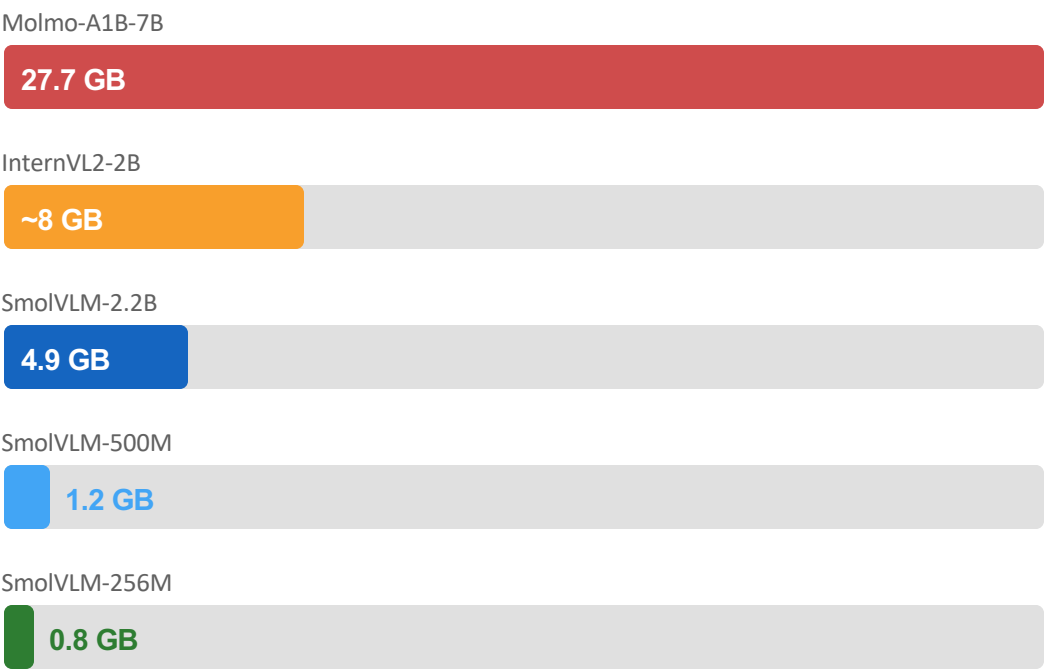
The Problem

- Leading VLMs (LLaVA, Idefics-80B) require massive compute and 27+ GB GPU memory
- Mobile / edge devices typically have <4 GB available for ML inference
- Cloud dependency adds latency, cost, and privacy concerns for multimodal tasks
- Simply shrinking large models loses accuracy; architectural rethinking is needed

Can we achieve competitive accuracy at <1 GB RAM?

SmolVLM answers yes — 256M params, 0.8 GB RAM, outperforming the 300× larger Idefics-80B.

GPU RAM Comparison

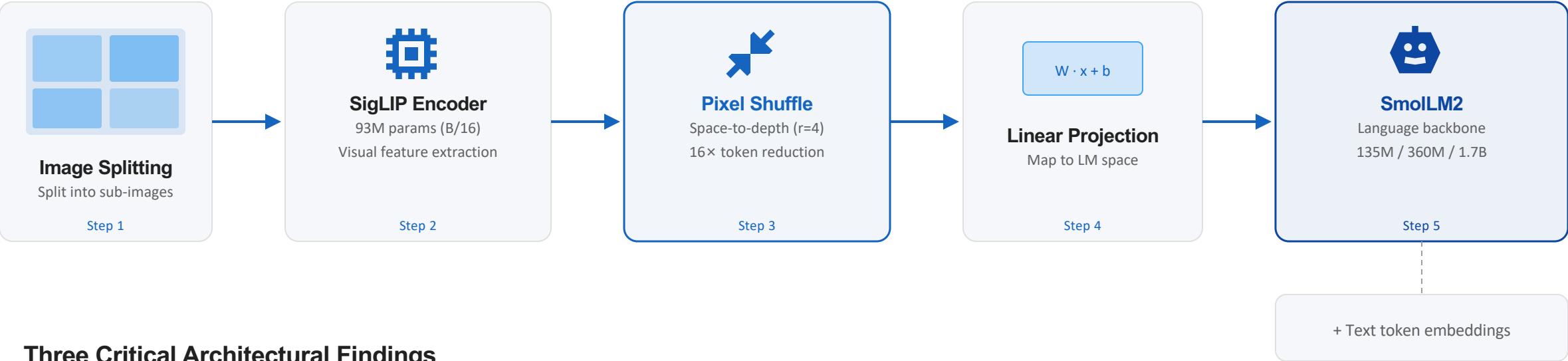


Scale: bar width proportional to RAM usage (max 27.7 GB)

35× less RAM

SmolVLM Architecture: Image Splitting → Pixel Shuffle → SmolLM2

A lightweight pipeline: split images into sub-images, encode with SigLIP, compress via pixel shuffle, and decode with SmolLM2.



Three Critical Architectural Findings

Finding 1: Balanced Compute
Smaller encoders (93M SigLIP) outperform larger ones (428M) for compact LMs
→ Slide 04

Finding 2: Extended Context
Scaling from 2K to 16K tokens via RoPE base adjustment (10K→273K)
→ Slide 06

Finding 3: Token Compression
Pixel shuffle $r=4$ gives 16× token reduction, optimal for small VLMs
→ Slide 05

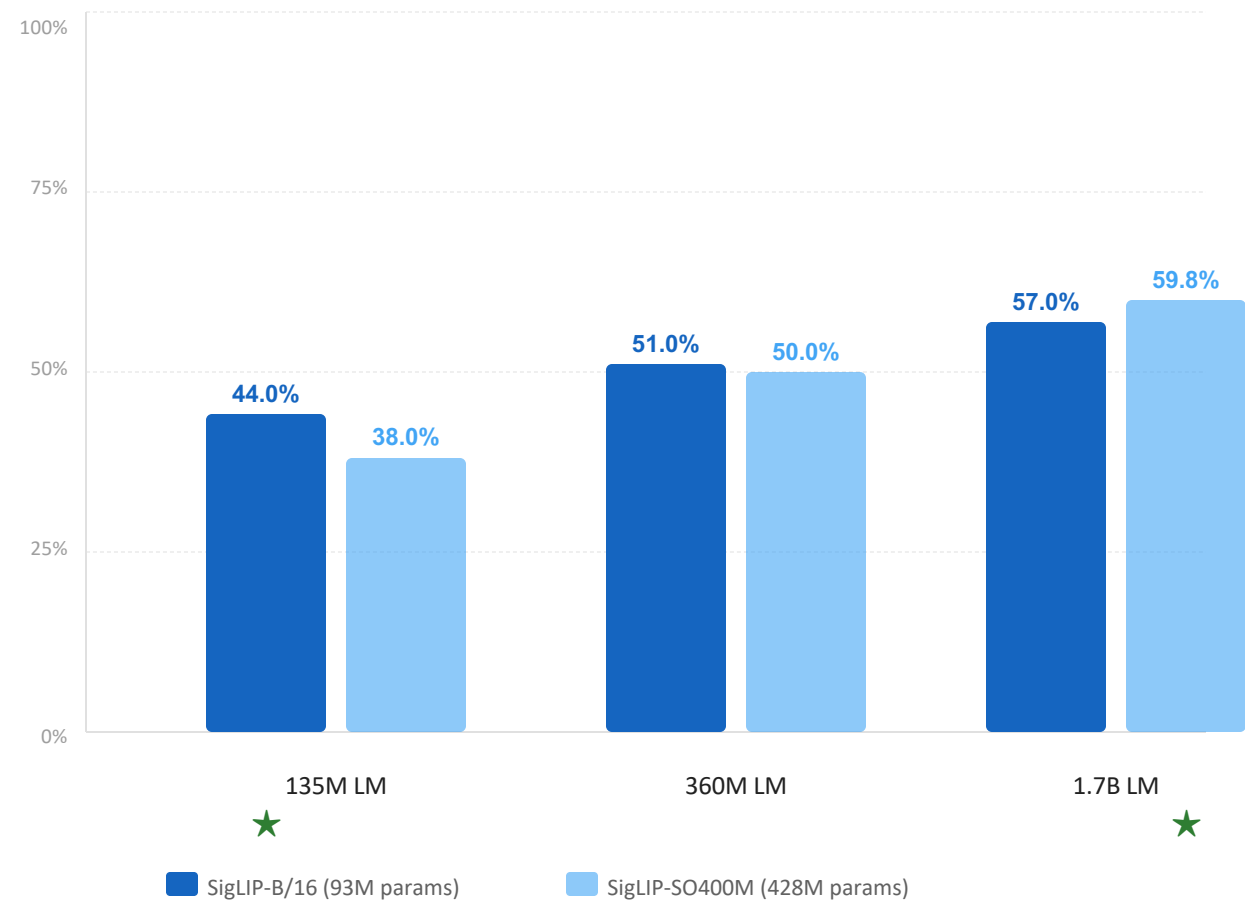
Video support: frames are sampled and processed through the same pipeline.

Smaller Vision Encoders Outperform Larger Ones in Compact VLMs

A 93M SigLIP-B/16 paired with a 135M LM outperforms a 428M SigLIP-SO400M — parameter balance matters more than encoder size.

Vision Encoder vs. LM Performance

Average benchmark score by encoder-LM combination



Key Insight

Encoder-LM compute balance matters

more than raw encoder size for small VLMs.

- At 135M LM: 93M encoder beats the 428M encoder (4.6× smaller)
- The 428M encoder adds 10% parameter overhead without benefit
- Only at 1.7B LM scale does the larger encoder justify its cost

Design Principle:

Match encoder capacity to LM capacity — don't over-invest in vision.

93M encoder
SigLIP-B/16 (selected)

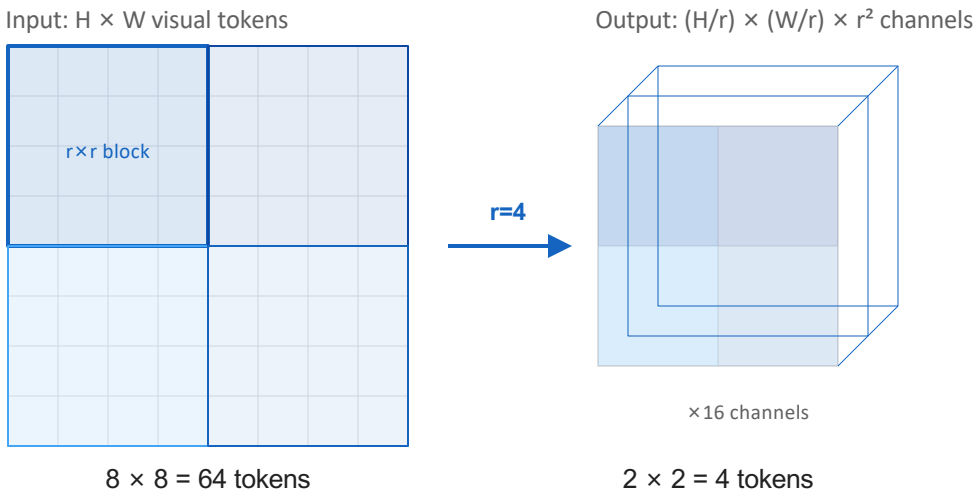
vs

428M encoder
SigLIP-SO400M (rejected)

Aggressive Pixel Shuffle (r=4) Cuts Visual Tokens 16×

Pixel shuffle rearranges spatial features into channel dimensions, dramatically reducing visual token count and attention overhead.

Pixel Shuffle Mechanism (Space-to-Depth)



64 → 4 tokens = 16× reduction

Transformation Formula

$(H, W, C) \rightarrow (H/r, W/r, C \times r^2)$ where r = shuffle factor

Pixel Shuffle Factor Comparison

Factor (r)	Token Reduction	Best For	Selected
$r = 2$	4×	Large models (2B+)	—
$r = 3$	9×	Medium models	—
$r = 4$	16×	Small models (<1B)	✓ SmolVLM

Benefits of $r=4$ for Small VLMs

- Reduces attention overhead ($O(n^2) \rightarrow$ much smaller n)
- Enables longer context (more sub-images per sequence)
- No significant accuracy loss for compact models
- Channel information preserves visual features

Extending Context from 2K to 16K Tokens Unlocks Higher Resolution

Performance improves significantly when context scales from 2K→16K, achieved by adjusting RoPE base from 10K to 273K.

 **The Limitation**

Default context: 2K tokens

- Only 1–2 sub-images fit per input
- Low-resolution image understanding
- Poor multi-image / video performance
- Standard RoPE base = 10K

 **The Solution**

Extended context: 8K–16K tokens

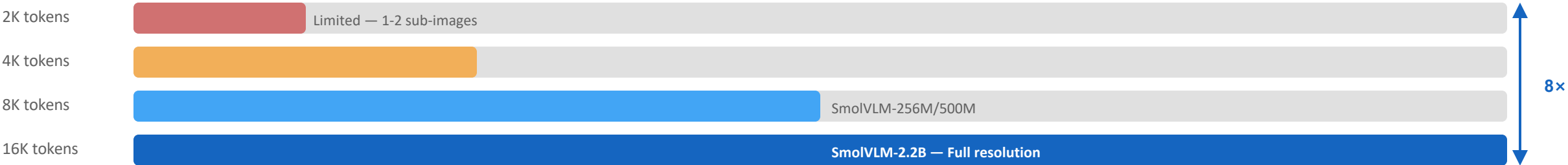
- RoPE base: 10K → 273K
- Fine-tune on mixed long/short data
- More sub-images per input
- Higher effective image resolution

 **Model Configuration**

Model	Context Length
SmolVLM-256M	8K tokens
SmolVLM-500M	8K tokens
SmolVLM-2.2B	16K tokens

RoPE base = 273,000

Context Length Scaling Impact



Longer context ↔ more sub-images ↔ higher effective resolution ↔ better accuracy

SmolVLM Achieves Competitive Performance at Sub-1GB to 4.9GB RAM

SmolVLM-256M outperforms the 300× larger Idefics-80B. The 2.2B model rivals models consuming 6× more RAM.

Capability	Benchmark	SmolVLM 256M	SmolVLM 500M	SmolVLM 2.2B	Best Efficient OS
		0.8 GB RAM	1.2 GB RAM	4.9 GB RAM	27.7 GB (Molmo) / ~8 GB (InternVL2)
Single-Image	OCRBench	52.6%	61.0%	72.9%	54.7% (Molmo-A1B-7B)
Single-Image	AI2D	46.4%	59.2%	70.0%	71.0% (Molmo-A1B-7B)
Single-Image	ChartQA	55.6%	62.8%	68.7%	48.0% (Molmo-A1B-7B)
Single-Image	TextVQA	50.2%	60.2%	73.0%	61.5% (Molmo-A1B-7B)
Single-Image	DocVQA	58.3%	70.5%	80.0%	77.7% (Molmo-A1B-7B)
Single-Image	ScienceQA	73.8%	80.0%	89.6%	87.5% (Molmo-A1B-7B)
Multi-task	MMMU	29.0%	33.7%	42.0%	33.9% (Molmo-A1B-7B)
Multi-task	MathVista	35.9%	40.1%	51.5%	37.6% (Molmo-A1B-7B)
Multi-task	MMStar	34.6%	38.3%	46.0%	43.1% (Molmo-A1B-7B)
Video	Video-MME	33.7%	42.2%	52.1%	45.0% (InternVL2-2B)
Video	MLVU	40.6%	47.3%	55.2%	48.2% (InternVL2-2B)
Average	All 11 Benchmarks	44.0%	51.0%	59.8%	—

SmolVLM-256M: 0.8 GB
Outperforms 300× larger Idefics-80B

SmolVLM-2.2B: 4.9 GB
Rivals SOTA at 6× less RAM

Molmo-A1B-7B: 27.7 GB
Best efficient open-source baseline

Real-Time Inference: Up to 80 Tokens/Second on Apple Silicon

HuggingSnap runs SmolVLM entirely on-device — no cloud, no latency, full privacy. Throughput scales linearly with batch size.



HuggingSnap iOS App

100% Local

No cloud dependency



256M on M1 Max 64GB

~80 tok/s

Batch size 1



256M at Batch 64 (high-end)

~1000 tok/s

Linear scaling with batch size



256M on MacBook Air M3 16GB

~55 tok/s

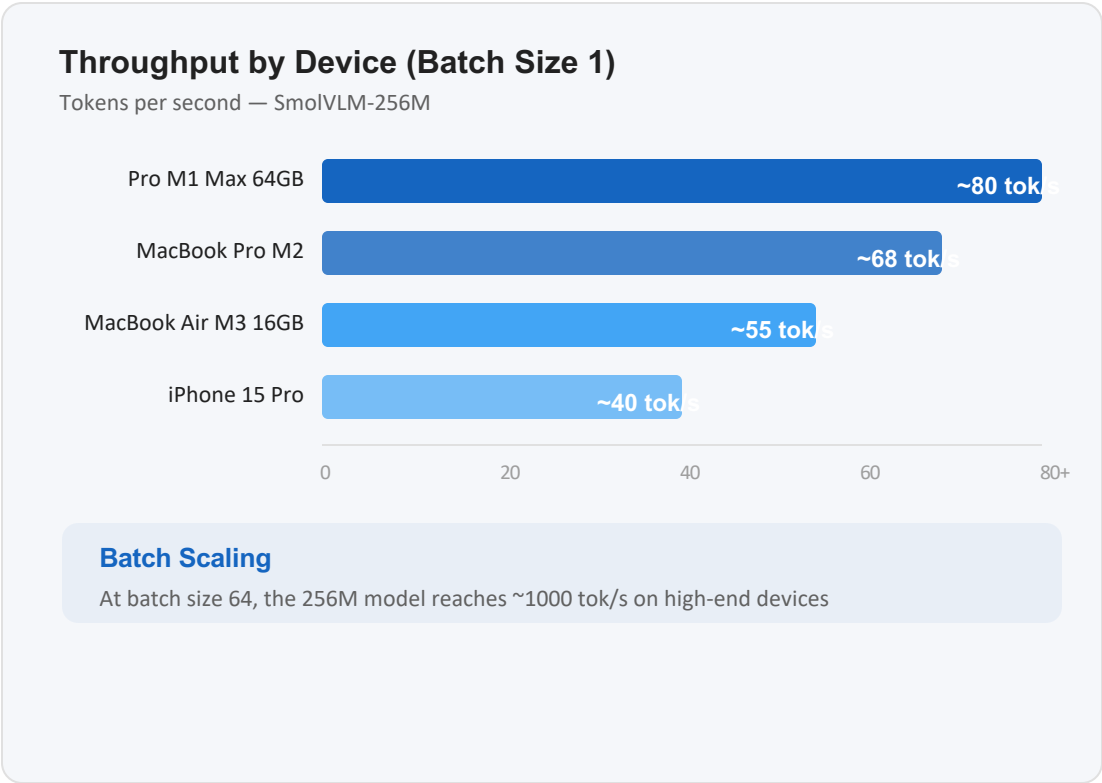
Consumer-grade hardware

Deployment Advantages

- Zero cloud latency — instant local inference
- Complete data privacy — no images leave the device
- Works offline — no network dependency



HuggingSnap is open-source — available on the iOS App Store



Key Takeaways: Design Principles for Efficient Multimodal Models

Five actionable design principles from SmolVLM — architectural efficiency matters more than raw scale.

1

Balance Encoder-LM Compute Allocation

A 93M SigLIP-B/16 outperforms the 428M SigLIP-SO400M for the smallest LMs. Match encoder capacity to LM capacity — don't over-invest in vision when the language model is the bottleneck.

2

Use Aggressive Token Compression for Small Models

Pixel shuffle at $r=4$ provides $16\times$ token reduction with no accuracy loss for compact VLMs. This eases attention overhead and enables longer-context modeling within the same memory budget.

3

Extend Context Length via RoPE Scaling

Scaling from 2K to 16K tokens (RoPE base $10K \rightarrow 273K$) unlocks higher image resolution and multi-image understanding. Fine-tune on mixed long/short-context data for robust generalization.

4

Sub-1GB Inference Is Achievable

SmolVLM-256M uses only 0.8 GB RAM and outperforms the $300\times$ larger LLaVA-80B. Enables deployment on smartphones and low-power edge devices with real-time throughput (~ 80 tok/s on Apple Silicon).

5

Fully Open-Source for Reproducibility

All model weights, training data, code, and the HuggingSnap mobile app are released as open source. Promotes reproducibility and accelerates community research in on-device multimodal AI.



Implication: Efficient architecture design — not scaling — is the path to democratizing multimodal AI on consumer hardware.